

It can handle many distributed transactions at once and give a more time-effective UI to work on such queries.

It is best for networking systems, and is also a native support tool for OpenTrace.

Benefits:

- Supports open tracing for popular languages such as Java, Python, Node, and so on.
- The modern user interface allows you to view a large amount of data
- Easy to deploy

The EFK stack

This uses the functionality of multiple tools like Elasticsearch, Kibana and Fluentd, which gives it a smooth functionality. It collects information through the pods working on cluster nodes, then stores the data in a central location, and in the end, the

visualisation of data is handled via Kibana.

It is easy to extract, search and display data through this tool.

Dynatrace

Dynatrace is highly adaptable with regard to whether you want to monitor from the application layer, down to the infrastructure level (provided you have access to the infrastructure layer). It provides full-stack monitoring and code level tracing.

The best part is that it is AI-powered and hence reduces the workload with regard to monitoring microservices involving a large network infrastructure. It delivers an answer with regard to the full stack, instantly.

Benefits:

- Can be used to monitor multiple OSs, whether physical or cloud based
- Offers complete dependency mapping and precise root cause determination

Depending on your needs, most of these tools are free. But if you want high-end monitoring resources, try Dynatrace. Kubernetes itself is a diverse platform. Hence, it is important for users to determine the time and money they are willing to spend on the system for a more streamlined rather than a manual monitoring tool. Maybe an organisation can spend more, but sticking to the basics and building up from there also works. **END** 🐧

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from one metre to one millimetre. Frequency varies from 300MHz to 300GHz. These are widely used for long-distance communication and are relatively less expensive. Devices using satellite technology allow users to stay connected wirelessly from anywhere on Earth.

Satellite communication consists of space and ground segments. The signal sent to the satellite through a device (satellite phone) is amplified by the satellite and sent back to the receiver antenna, which is located on Earth.

Efforts are on to leverage the benefits of technology for the betterment of mankind. Important initiatives pursued by Indian Space Research Organisation towards societal development include tele-education, tele-medicine, village resource centre and disaster management system programmes. The potential of space technology for applications of national development and television broadcast is enormous.

Fixed satellite services support international Internet connectivity to private business networks. Mobile satellite services that use a variety of transportable receiver and transmitter equipment provide communication services for land mobile, maritime and aeronautical customers. Broadcast satellite services (BSS) offer high transmission power for reception using very small ground equipment.

LoRaWAN

Long Range (LoRa) technology is designed to wirelessly connect battery-operated things to the Internet in regional, national or global networks. It has large potential in smart cities, smart homes and buildings, smart agriculture, smart metering, smart supply chain and logistics, and many other applications.

LoRa connects devices up to fifteen kilometres apart in rural areas, and

penetrates dense urban or deep indoor environments. This requires minimum energy, with a prolonged battery lifetime of up to ten years, minimising battery replacement costs.

LoRa features end-to-end AES128 encryption, mutual authentication, integrity protection and confidentiality. LoRa modulation is based on Chirp spread-spectrum technology, which makes it perform well with multipath fading, channel noise and Doppler effect, even at low power. LoRaWAN for wide-area networks can use channels with a bandwidth of 125kHz, 250kHz or 500kHz, depending on the region or frequency plan. **END** 🐧

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